PHYSICS 2111 · PHYSICS NINJA PRACTICE EXAM

Exam 2 · Version B

20 questions · 90 minutes · original practice exam, not an official exam

Name _____ NetID _____ Section _____

Closed book. Calculators allowed. Use the official PHY 2111 formula sheet. Take $g = 9.81\ \text{m/s}^2$.
MC5 questions have five choices (6 points), MC3 three choices (3 points), MC2 two choices (2 points).
Mark one answer per question. Answers and worked solutions are in the separate solutions file.

THE NEXT 2 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A 4 kg block is released from rest at the top of a ramp 3 m high. On the way down, friction does −40 J of work on it.

1) What is the speed of the block at the bottom of the ramp? **MC5**

a. 6.23 m/s
b. 7.67 m/s
c. 8.88 m/s
d. 4.47 m/s
e. 4.41 m/s

2) The change in the mechanical energy of the block is **MC3**

a. +40 J
b. 0
c. −40 J

THE NEXT 3 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A 2 kg block slides at 3 m/s across a frictionless floor and runs into a spring with k = 450 N/m.

3) What is the maximum compression of the spring? **MC5**

a. 0.0133 m
b. 0.04 m
c. 0.283 m
d. 0.141 m
e. 0.2 m

4) What is the speed of the block when the spring is compressed 0.10 m? **MC5**

a. 1.5 m/s
b. 2.12 m/s
c. 2.6 m/s
d. 3 m/s
e. 2.25 m/s

5) The magnitude of the block's acceleration is largest **MC3**

a. halfway to maximum compression.
b. at the moment it first touches the spring.
c. at maximum compression.

THE NEXT 2 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A force acts on a 3.0 kg cart along the x axis. It rises linearly from 0 at x = 0 to 30 N at x = 4.0 m, then stays at 30 N until x = 6.0 m. The cart starts from rest on a frictionless track.

6) How much work does the force do from x = 0 to x = 6.0 m? **MC5**

a. 90 J
b. 120 J
c. 60 J
d. 150 J
e. 180 J

7) What is the speed of the cart at x = 6.0 m? **MC5**

a. 6.32 m/s
b. 8.94 m/s
c. 12.6 m/s
d. 11 m/s
e. 7.75 m/s

THE NEXT 2 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A 60 kg skier leaves the top of a 40 m high frictionless hill moving at 5.0 m/s.

8) What is the skier's speed at the bottom? **MC5**

a. 23 m/s
b. 28 m/s
c. 33 m/s
d. 28.3 m/s
e. 28.5 m/s

9) A heavier skier starting the same way would reach the bottom with a speed that is **MC3**

a. smaller.
b. larger.
c. the same.

THE NEXT 2 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

An 80 kg astronaut floating at rest outside a space station throws a 4.0 kg wrench at 6.0 m/s.

10) What is the astronaut's recoil speed? **MC5**

- **a.** 120 m/s
- **b.** 6 m/s
- **c.** 0.05 m/s
- **d.** 0.3 m/s
- **e.** 0.316 m/s

11) After the throw, the kinetic energy of the wrench is **MC3**

- **a.** less than the astronaut's.
- **b.** equal to the astronaut's.
- **c.** greater than the astronaut's.

THE NEXT 2 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A 1000 kg car moving east at 15 m/s collides at an intersection with a 1500 kg car moving north at 10 m/s. The cars stick together.

12) What is the speed of the wreck just after the collision? **MC5**

- **a.** 25 m/s
- **b.** 8.49 m/s
- **c.** 18 m/s
- **d.** 6 m/s
- **e.** 10 m/s

13) In what direction does the wreck move just after the collision? **MC5**

- **a.** 56.3° north of east
- **b.** due east
- **c.** 33.7° north of east
- **d.** 45.0° north of east
- **e.** due north

THE NEXT 2 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

Three small objects lie on the x axis: 2.0 kg at x = 0, 3.0 kg at x = 2.0 m and 5.0 kg at x = 4.0 m.

14) Where is the center of mass? **MC5**

- **a.** 3 m
- **b.** 2.67 m
- **c.** 2 m
- **d.** 1.3 m
- **e.** 2.6 m

15) The objects sit on frictionless ice and push each other apart. The center of mass **MC2**

a. moves toward the 5.0 kg object.
b. stays where it is.

THE NEXT 2 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A 0.020 kg bullet is fired into a 2.0 kg wooden block hanging from a long string. The bullet embeds in the block, and the block swings up 0.12 m.

16) What was the bullet's speed just before impact? **MC5**

a. 153 m/s
b. 155 m/s
c. 1.53 m/s
d. 15.5 m/s
e. 1550 m/s

17) During the very short collision itself, which quantity is conserved? **MC3**

a. Momentum only
b. Kinetic energy only
c. Both momentum and kinetic energy

THE NEXT QUESTION PERTAINS TO THE SITUATION DESCRIBED BELOW.

A car moves at 72 km/h. Its tires have a radius of 0.35 m and roll without slipping.

18) What is the angular speed of the tires? **MC5**

a. 9.09 rad/s
b. 57.1 rad/s
c. 7 rad/s
d. 206 rad/s
e. 28.6 rad/s

THE NEXT 2 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A solid disk of mass 8.0 kg and radius 0.30 m spins at 120 rpm about its central axis.

19) What is its rotational kinetic energy? **MC5**

a. 28.4 J
b. 4.52 J
c. 14.2 J
d. 56.8 J
e. 2590 J

20) A thin ring with the same mass and radius spinning at the same rate has a kinetic energy that is MC3

a. twice as large.

b. half as large.

c. the same.
